Design and control of the Six-DOF robotic manipulator

Authors: Ruohan He, Yinrou Wang, Tingyi Li (2024)

ABSTRACT

This paper presents the systematic mechanical design and simulation of a 6-DOF robotic manipulator. A CAD model is created in SOLIDWORKS and analyzed using Finite Element Analysis (FEA) to evaluate stress and vibrations. The model is then imported into MATLAB/Simulink to simulate coordinate transformations, trajectory tracking, and verify joint angle/torque limits.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Serves as a direct reference for the mechanical structural validation and joint torque limit settings of the EB15 3D-printed model.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Ruohan He provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.